SCT3258 Introduction

SCT3258 is a low power high performance base band processor supporting DPMR and DMR protocol. With an external codec (audio band ADC and DAC), it completes the entire physical layer and data link layer, and most of the call control layer processing of DPMR and DMR protocol. In DPMR mode, SCT3258 complies with ETSI TS 102 490 and ETSI TS 102 658. In DMR mode, it complies with ETSI TS 102 361. Figure 1-1 shows SCT3258 used in a DPMR or DMR system with conventional RF circuits. Figure 1-2 shows SCT3258 used with SCT3700. SCT3258 is designed for easy migration from analog radios. The system designer can start with a typical analog radio, replacing the analog base band processor with SCT3258 together with an external codec, to obtain the simplest form of a DPMR or DMR radio. To achieve the full potential of DPMR or DMR, application level software programming is required on the host processor.

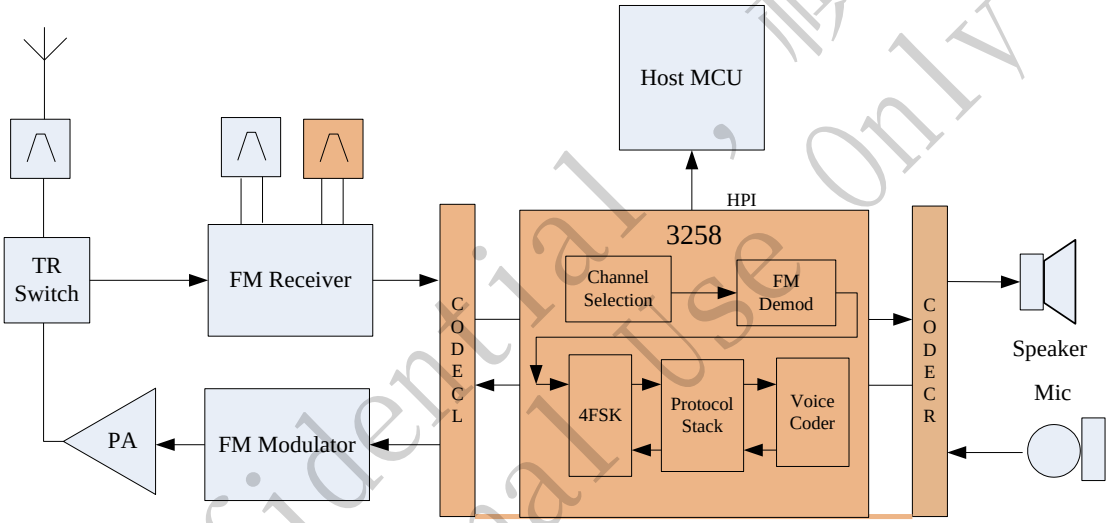


Figure 1-1 DPMR / DMR System Diagram with Conventional RF

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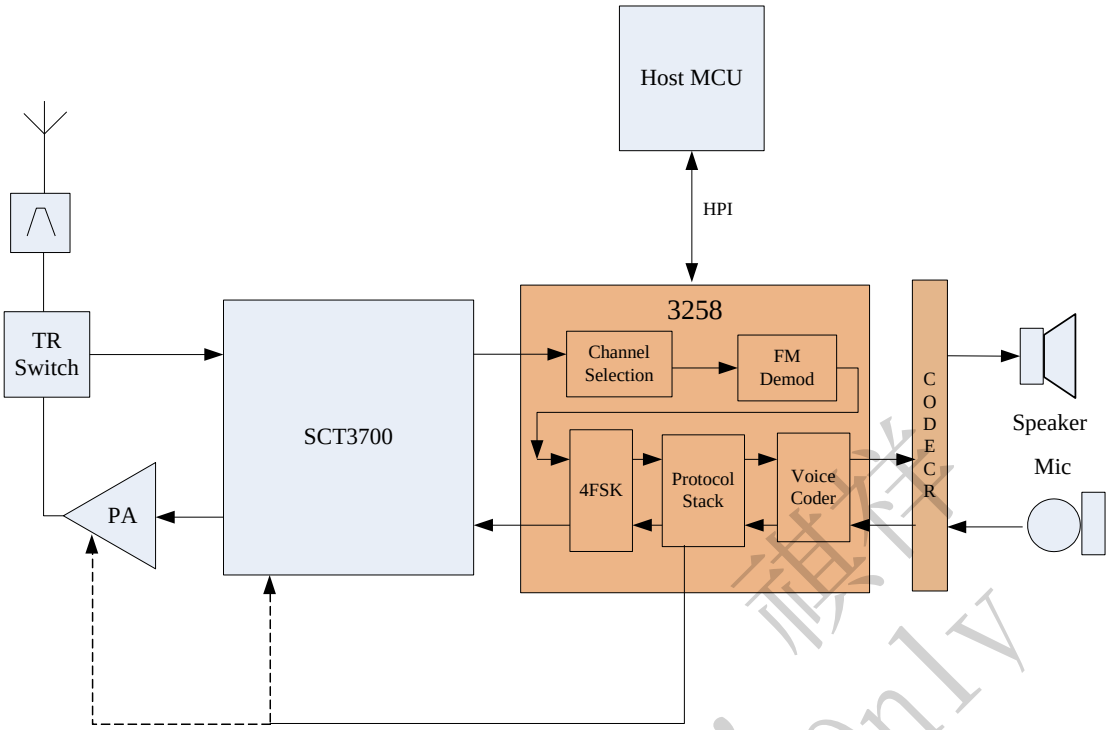


Figure 1-2 DPMR / DMR System Diagram with SCT3700

SCT3258 contains two high-speed serial ports, a single 8-bit processor interface (HPI), and a 4-pin programmable I/O port. The two serial ports are used for connection with external codec and external vocoder.

SCT3258 communicates with host processor with 8 bit HPI port (Intel Mode or Motorola Mode).

The firmware of SCT3258 is stored on chip. At the start of the system, the host processor downloads a small boot loader to SCT3258 to get SCT3258 started. SCT3258 then loads the entire firmware and starts execution.

SCT3258 core logic operates at 1.2 V, and the I/O operates at 2.5-3.3V.

SCT3258 is packaged in QFN64

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Features

DPMR

- Support DPMR Tier 1 (ETSI TS 102 490)
- Support DPMR Tier 2 (ETSI TS 102 658) Mode 1 and Mode 2
- Air interface physical layer (layer 1)
- Air interface data link layer (layer 2)
- Air interface call control layer (layer 3)
- Full Annex A support, with BCD addressing and automatic call match

DMR

1 Support DMR Tier1 and Tier 2 (ETSI TS 102 361)
2 Air interface physical layer (layer 1)
3 Air interface data link layer (layer 2)
4 Air interface call control layer (layer 3)
5 Annex C (TS 102 361-2 Annex C) support, with BCD addressing and automatic call match
6 Transmit in slotted or continuous mode
7 Receive in slotted or continuous mode

4 FSK Modem

- 4800 bps data rate for DPMR and 9600 bps for DMR
- Automatic frame sync detection
- Programmable modulation index
- Support two point modulation, and I/Q modulation
- BER Test Mode complied with ITU O.153

Vocoder

- Build-in ASELP vocoder from Tsinghua University (with 2mm x 1mm security key)
- Build-in AMBE + 2C vocoder from DVSI (no security key)
- Build-in AMBE + 2 vocoder from DVSI (no security key, soon to come)
- External vocoder interface for AMBE3000
- Support other types of low bit rate vocoder with 3600 bps
- Support 1031 Hz Tone and Silence Test Mode
- Automatic vocoder switching at the receiver in DPMR mode

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Analog Mode Support

- Support voice channel filters (LPF/HPF/Limiter), as well as pre-emphasis and de-emphasis filters.
- Support CTCSS/DCS generation and detection
- Support arbitrary CTCSS/DCS code, and blind detection
- Support compander
- Automatic mode detection (analog or DPMR) in receiver mode

DPMR Premium Features

- Voice Recording and Playing back for local and remote
- 16 / 64 / 192 bit voice encryption

I/O

- Two high-speed TDM serial ports
- 8-bit HPI
- 4-pin PIO port

PLL

- On chip PLL circuit which provide system clock of up to 110.592MHz.

Technology

- QFN64 package
- Low power process, 38 mW in DPMR mode and 64 mW in DMR mod

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